

# Log-concavity of reduced composition polynomials via adjacent B-spline refinement columns

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## Abstract

Ardila and Docker associated a reduced composition polynomial  $f_c(q)$  with every positive integer composition  $c$ . They proved positivity of its coefficients and asked whether those coefficients are unimodal or, more strongly, log-concave. We prove the stronger assertion for every composition. The first ingredient identifies the coefficient vector, up to a positive scalar, with a column of a B-spline knot-refinement matrix. Total nonnegativity, constant reproduction, and variation diminution then give unimodality. For log-concavity, deleting the first or last endpoint of the composition produces two adjacent columns of a lower-degree refinement matrix. Their adjacent  $2 \times 2$  minors are nonnegative. An exact identity converts those minors into the log-concavity defects of  $f_c$ , with the remaining sign case closed by unimodality. Thus both the unimodality question and the stronger log-concavity question have affirmative answers.

## 1 Introduction

Let  $c = (d_1, \dots, d_r)$  be a composition of  $n$  into positive parts. Ardila and Docker introduced a composition polynomial  $g_c(q)$  in their study of subdivisions of lifted generalized permutahedra [1]. They showed that

$$g_c(q) = (1 - q)^r f_c(q),$$

where the reduced composition polynomial  $f_c(q)$  has positive coefficients. Their Question 7.5 asks whether these coefficients are unimodal; immediately afterwards they ask whether the stronger property of log-concavity always holds. The original paper verified the expected shape in a large finite family but left both questions open.

Novelli and Thibon later gave a combinatorial interpretation of  $n!f_c(q)$  as a generating polynomial for a statistic on permutations [4]. That interpretation explains coefficient integrality and positivity for the scaled polynomial  $n!f_c(q)$ , but it does not by itself provide the required second-order coefficient inequalities.

Our proof uses a different interface. The explicit coefficient formula for  $f_c$  is a rising-factorial divided difference. This is exactly the formula for a discrete B-spline, or equivalently for one column of a B-spline knot-refinement matrix [2]. The one-column consequence is unimodality.

The stronger result comes from looking at two columns at once: the reduced polynomials obtained by deleting the final and initial parts of  $c$  form adjacent refinement columns, so total nonnegativity controls their adjacent determinants. A short identity turns this determinant inequality into log-concavity of the original coefficient sequence.

**Theorem 1.1.** *For every positive integer composition  $c$ , the coefficient sequence of the reduced composition polynomial  $f_c(q)$  is log-concave. In particular, it is unimodal.*

Here a finite nonnegative sequence  $(a_i)$  is log-concave when

$$a_i^2 \geq a_{i-1}a_{i+1}$$

at every internal index. All coefficient sequences below are extended by zero outside their natural support.

The spline results used here are classical. The contribution is the exact identification of the composition-polynomial coefficients with refinement columns and the adjacent-column determinant argument that proves Theorem 1.1. In particular, the log-concavity of a continuous B-spline as a function on its support [3] is distinct from the discrete coefficient log-concavity proved here.

## 2 Reduced composition polynomials and discrete B-splines

Let

$$c = (d_1, \dots, d_r), \quad n = d_1 + \dots + d_r,$$

and write its partial sums as

$$0 = \beta_0 < \beta_1 < \dots < \beta_r = n.$$

Write

$$f_c(q) = \sum_{i=0}^{n-r} F_i q^i.$$

Ardila and Doker's coefficient formula is

$$F_i = \sum_{\substack{0 \leq j \leq r \\ \beta_j \leq i}} \frac{(-1)^j}{[\beta_j]} \binom{r+i-\beta_j-1}{r-1}, \quad [\beta_j] = \prod_{\ell \neq j} |\beta_\ell - \beta_j|. \quad (2.1)$$

We now place (2.1) in the discrete-spline setting. For strictly increasing integer knots  $z_0 < \dots < z_s$ , define the unnormalized discrete B-spline of degree  $s-1$  and unit mesh by

$$U^{s-1}(x; z_0, \dots, z_s) = (-1)^s [(x - \cdot)^{\overline{s-1}} \mathbf{1}_{x > \cdot}] [z_0, \dots, z_s], \quad (2.2)$$

where  $y^{\overline{m}} = y(y+1)\dots(y+m-1)$  and the final brackets denote a divided difference in the knot variable. Its normalized version is

$$V^{s-1}(x; z_0, \dots, z_s) = (z_s - z_0) U^{s-1}(x; z_0, \dots, z_s). \quad (2.3)$$

**Lemma 2.1.** For  $0 \leq i \leq n - r$ ,

$$V^{r-1}(i+1; \beta_0, \dots, \beta_r) = n(r-1)!F_i. \quad (2.4)$$

The values on the left vanish on the surrounding integer grid outside this index range.

*Proof.* The divided-difference denominator at  $\beta_j$  has sign  $(-1)^{r-j}$  and absolute value  $[\beta_j]$ . Moreover,

$$\frac{(i+1-\beta_j)^{\overline{r-1}}}{(r-1)!} = \binom{r+i-\beta_j-1}{r-1}.$$

Substituting the knots  $\beta_0, \dots, \beta_r$  into (2.2), and then using (2.3), gives (2.4) term by term from (2.1).

It remains to verify that the divided difference in (2.4) is a refinement coefficient, rather than merely a formally similar discrete-spline value. Embed the local knot window in

$$\tau = (-r, \dots, -1, \beta_0, \dots, \beta_r, n+1, \dots, n+r)$$

and refine to the unit knot vector

$$t = (-r, -r+1, \dots, n+r).$$

At a fine-grid left endpoint  $i$ , the refinement formula is the divided difference of a truncated polynomial kernel. More precisely, Theorem 1 and equations (3.20)–(3.23) of Cohen–Lyche–Riesenfeld [2], specialized to degree  $r-1$ , unit fine mesh, and the selected coarse window, give its refinement coefficient as

$$\alpha_i = n[\beta_0, \dots, \beta_r] \left( \mathbf{1}_{y>i} \prod_{h=1}^{r-1} (y-i-h) \right), \quad (2.5)$$

where the divided difference is taken in  $y$ . Put  $x = i+1$  and

$$R(y) = (x-y)^{\overline{r-1}}.$$

At every integer knot,

$$\prod_{h=1}^{r-1} (y-i-h) = (-1)^{r-1} R(y). \quad (2.6)$$

The polynomial  $R$  has degree  $r-1$ , so its order- $r$  divided difference on the  $r+1$  knots is zero. On integer knots,  $y > i$  and  $x > y$  are complementary conditions. When  $r > 1$ , the possible boundary value  $y = x$  vanishes because  $R(x) = 0$ ; when  $r = 1$ , strict complementarity holds directly. Therefore

$$[\beta_0, \dots, \beta_r](\mathbf{1}_{y>i} R) = -[\beta_0, \dots, \beta_r](\mathbf{1}_{x>y} R). \quad (2.7)$$

Substitution of (2.6)–(2.7) into (2.5) gives

$$\alpha_i = V^{r-1}(i+1; \beta_0, \dots, \beta_r),$$

including its sign and normalization. Finally, if  $i < 0$ , the truncated kernel in (2.5) agrees at all coarse knots with one polynomial of degree  $r-1$ , so its order- $r$  divided difference vanishes. If  $i > n-r$ , every potentially nonzero term contains a zero factor. Hence the nonzero core has precisely the indices in (2.4).  $\square$

### 3 Knot refinement and unimodality

We recall only the refinement properties needed below. Suppose a coarse knot vector is refined by successive knot insertions, without changing the spline degree. If  $p$  is the coarse control vector and  $d$  the refined control vector, write

$$d = Kp.$$

For one knot insertion, every new control coefficient is either copied from an old coefficient or is a convex combination of two adjacent old coefficients. In the convention above, the rows of  $K$  are fine indices and its columns are coarse indices. A one-step matrix is therefore a nonnegative bidiagonal matrix with row sums one. It is totally nonnegative, and products preserve total nonnegativity and row sums.

For a real vector  $v$ , let  $S^-(v)$  be the number of sign changes after its zero entries are deleted. We use the standard variation-diminishing theorem for totally nonnegative matrices [5, Chapter 3]:

$$S^-(Kv) \leq S^-(v). \tag{3.1}$$

**Lemma 3.1.** *Every column of a finite B-spline knot-refinement matrix, with zeros retained outside its local support, is unimodal.*

*Proof.* Let  $\alpha = Ke_s$  be a column and choose  $0 < \lambda < 1$ . Since  $K\mathbf{1} = \mathbf{1}$ ,

$$\alpha - \lambda\mathbf{1} = K(e_s - \lambda\mathbf{1}). \tag{3.2}$$

The vector on the right before applying  $K$  has sign pattern

$$- \cdots - + - \cdots -,$$

so it has at most two sign changes. Equation (3.1) gives the same upper bound for the left side of (3.2).

If  $\alpha$  were not unimodal, two higher values would be separated by a strictly lower value. Choose  $\lambda$  strictly between the valley and the smaller of the two higher values. Including the zero entries on the two sides of the local support,  $\alpha - \lambda\mathbf{1}$  would then have a subpattern

$$-, +, -, +, -,$$

and hence at least four sign changes, a contradiction.  $\square$

**Corollary 3.2.** *For every positive integer composition  $c$ , the coefficient sequence of  $f_c(q)$  is unimodal.*

*Proof.* By Lemma 2.1, the coefficient sequence is a positive scalar multiple of one refinement column. Apply Lemma 3.1.  $\square$

Thus the first part of Ardila–Doker Question 7.5 follows from a one-column shape property. To reach log-concavity we use two neighboring columns.

## 4 Adjacent endpoint-deletion columns

Assume for now that  $r \geq 2$ . Define

$$A(q) = f_{(d_1, \dots, d_{r-1})}(q), \quad B(q) = q^{d_1} f_{(d_2, \dots, d_r)}(q), \quad (4.1)$$

and write their zero-padded coefficients as  $A_i$  and  $B_i$ . The shift in  $B(q)$  places both sequences on the global unit grid determined by  $0, 1, \dots, n$ .

The two knot windows

$$(\beta_0, \dots, \beta_{r-1}) \quad \text{and} \quad (\beta_1, \dots, \beta_r) \quad (4.2)$$

are adjacent windows in a common degree- $(r-2)$  coarse B-spline basis. Refine the padded coarse knot vector

$$\tau = (-(r-1), \dots, -1, \beta_0, \dots, \beta_r, n+1, \dots, n+r-1)$$

to the unit grid

$$t = (-(r-1), -(r-2), \dots, n+r-1).$$

Lemma 2.1, applied to the two shorter compositions, identifies the corresponding adjacent columns as

$$\alpha_i^L = \beta_{r-1}(r-2)!A_i, \quad \alpha_i^R = (n-\beta_1)(r-2)!B_i. \quad (4.3)$$

Both factors are strictly positive. The refinement matrix is totally nonnegative, and its rows and columns in (4.3) are in increasing order. Consequently every adjacent  $2 \times 2$  minor gives

$$M_i := A_i B_{i+1} - A_{i+1} B_i \geq 0. \quad (4.4)$$

The orientation in (4.4) is important: rows are consecutive fine-grid indices, and the left window in (4.2) precedes the right window.

## 5 The determinant identity

Continue to write

$$F(q) = f_c(q) = \sum_i F_i q^i,$$

with  $F_i = 0$  outside the natural support, and set

$$\Delta_i = F_i - F_{i-1}, \quad L_i = F_i^2 - F_{i-1}F_{i+1}. \quad (5.1)$$

The quantity  $L_i$  is the log-concavity defect at  $i$ .

**Lemma 5.1.** *The endpoint-deletion polynomials satisfy*

$$A(q) - B(q) = n(1 - q)F(q) \quad (5.2)$$

and

$$B(q) = q(rF(q) - (1 - q)F'(q)). \quad (5.3)$$

Equivalently, with  $k = r - 1$ ,

$$A_i - B_i = n\Delta_i, \quad B_i = (k + i)F_{i-1} - iF_i. \quad (5.4)$$

*Proof.* The explicit partial-fraction formula of Ardila–Doker is

$$g_c(q) = \sum_{j=0}^r \frac{(-1)^j q^{\beta_j}}{[\beta_j]}. \quad (5.5)$$

Its endpoint-deletion formulas (Corollary 7.3 of [1]) read

$$g_{(d_1, \dots, d_{r-1})}(q) = \sum_{j=0}^r \frac{(-1)^j (n - \beta_j) q^{\beta_j}}{[\beta_j]}, \quad (5.6)$$

and

$$q^{d_1} g_{(d_2, \dots, d_r)}(q) = - \sum_{j=0}^r \frac{(-1)^j \beta_j q^{\beta_j}}{[\beta_j]}. \quad (5.7)$$

Adding the right side of (5.6) to the negative of the right side of (5.7) gives  $ng_c(q)$ . Since  $g_c = (1 - q)^r F$ , while the two truncated polynomials in (5.6)–(5.7) have  $r - 1$  parts, division by  $(1 - q)^{r-1}$  gives (5.2).

Equations (5.7) and (5.5) also give

$$q^{d_1} g_{(d_2, \dots, d_r)}(q) = -qg'_c(q).$$

Differentiating  $g_c(q) = (1 - q)^r F(q)$  therefore gives

$$(1 - q)^{r-1} B(q) = q(1 - q)^{r-1} (rF(q) - (1 - q)F'(q)),$$

which proves (5.3). Extracting coefficients from (5.2) and (5.3) yields (5.4).  $\square$

**Lemma 5.2.** *For every integer  $i$ ,*

$$M_i = n(kL_i - \Delta_i \Delta_{i+1}), \quad k = r - 1. \quad (5.8)$$

*Proof.* Using  $A_i = B_i + n\Delta_i$  in (4.4) gives

$$M_i = n(\Delta_i B_{i+1} - \Delta_{i+1} B_i).$$

Substitute the second identity in (5.4), once at  $i$  and once at  $i + 1$ . Expanding and collecting the coefficients of  $F_{i-1}F_i$ ,  $F_i^2$ , and  $F_iF_{i+1}$  yields

$$\Delta_i B_{i+1} - \Delta_{i+1} B_i = k(F_i^2 - F_{i-1}F_{i+1}) - (F_i - F_{i-1})(F_{i+1} - F_i),$$

which is (5.8). □

## 6 Proof of the main theorem

We first treat  $r \geq 2$ . By Corollary 3.2,  $(F_i)$  is a nonnegative unimodal sequence. Hence its first differences can change sign only from nonnegative to nonpositive; a negative difference can never be followed later by a positive one.

Fix an internal index  $i$ . If

$$\Delta_i \Delta_{i+1} \geq 0,$$

then (4.4) and (5.8) imply

$$kL_i \geq \Delta_i \Delta_{i+1} \geq 0.$$

Because  $k = r - 1 > 0$ , this gives  $L_i \geq 0$ .

It remains to consider  $\Delta_i \Delta_{i+1} < 0$ . Unimodality excludes  $\Delta_i < 0 < \Delta_{i+1}$ , so necessarily

$$\Delta_i > 0 > \Delta_{i+1}.$$

Thus  $F_i$  is strictly larger than both neighboring coefficients. Since all coefficients are nonnegative,

$$F_i^2 > F_{i-1}F_{i+1},$$

and again  $L_i \geq 0$ .

When  $r = 1$ , one has directly

$$f_{(n)}(q) = \frac{1 + q + \cdots + q^{n-1}}{n},$$

whose coefficients are log-concave. If  $r \geq 2$  but  $n - r \leq 1$ , the support has at most two terms and there is no internal inequality to check. With zero padding, the two endpoint inequalities are also immediate. This proves Theorem 1.1 for every positive integer composition. □

## 7 Examples and scope

For  $n = 4$ , scaled reduced composition polynomials listed by Novelli–Thibon include

$$4!f_{(4)}(q) = 6 + 6q + 6q^2 + 6q^3,$$

$$4!f_{(3,1)}(q) = 2 + 4q + 6q^2,$$

and

$$4!f_{(2,2)}(q) = 3 + 6q + 3q^2.$$

The first is flat, the second is increasing, and the third has an interior peak; all three are log-concave. The proof above treats these patterns uniformly. It also permits equality in the log-concavity inequalities, so no strictness assertion is intended.

The result concerns the coefficient sequence of  $f_c(q)$ . It does not assert real-rootedness, ultra-log-concavity, or a Pólya-frequency property. Nor does it claim that arbitrary columns of an arbitrary nonnegative or stochastic matrix are log-concave. The determinant argument uses the specific pair of adjacent endpoint-deletion columns together with the exact identity (5.8).

## 8 Computational verification

The proof is symbolic and does not depend on finite computation. The exact-arithmetic script `test_truncation_pencil.py` uses (2.1) to construct the coefficients; it does not independently prove that source formula. For all 8,178 nontrivial compositions of total size at most 13, it checked:

- the coefficient form of (5.2), including the zero-padded endpoint shifts;
- 53,157 adjacent-column determinants  $M_i \geq 0$ ;
- the same 53,157 instances of identity (5.8);
- 36,813 internal log-concavity defects.

No failure occurred. This finite calculation is an algebraic regression test, not a substitute for the general proof.

The separate script `verify_composition_unimodality.py` checked positivity, unimodality, and log-concavity in 75,535 test instances, representing 74,913 distinct compositions. It also compared the explicit coefficients with actual SciPy knot insertions on 199 distinct fixed-seed samples and included deliberately shifted and reversed negative controls. Those checks support the implementation of Lemma 2.1 but are likewise not used as a logical premise of Theorem 1.1.

## 9 Relation to prior work

Ardila–Doker [1] introduced the polynomials, proved coefficient positivity, and posed unimodality and log-concavity as questions. Their Corollary 9.2 also proves log-concavity of



the linear-extension numbers  $N_j$  in a Bernstein-basis expansion of  $g_c(q)$ . That is a different coefficient sequence in a different basis from the power-basis coefficients  $F_i$  of  $f_c(q)$  treated here. Novelli–Thibon [4] supplied a combinatorial interpretation through permutations and noncommutative symmetric functions. Cohen–Lyche–Riesenfeld [2] developed the discrete B-spline refinement formulas and subdivision machinery used in Lemma 2.1. Floater [3] proved log-concavity of continuous B-spline functions; that theorem concerns function values on a continuous support rather than the discrete refinement coefficients considered here. The variation-diminishing implication used in Section 3 is the standard total-positivity result recorded, for example, by Pinkus [5].

The one-column refinement argument proves unimodality. The two-column argument then strictly strengthens it to log-concavity. These are therefore presented as one result rather than as two independent contributions.

A targeted literature refresh through September 3, 2026 found no publication directly proving log-concavity for all reduced composition polynomials. This is a bounded novelty search, not a claim that all unindexed or unpublished work has been excluded.

## AI-assisted research and writing disclosure

The author used OpenAI Codex as an AI-assisted tool for literature organization, proof exploration, exact verification, adversarial auditing, and manuscript drafting. Carpotopus is responsible for the content of the manuscript.

## References

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